

Gate A review - Marketplace Seller Orders

1. Unlock condition

This file is protected answer material. Open it only after the Gate SQL file and evidence have been saved. Do not use it to construct the first attempt.

2. What a sound solution should do

Work only with the single table `seller_orders`. State source grain and each changed result grain. Use `WHERE` for source-row filtering, `GROUP BY` for summary buckets, `HAVING` for group-level filters, `ORDER BY` plus `LIMIT` for deterministic top-N, and `CASE` with the assigned thresholds. No `JOIN`, `CTE`, subquery or window function is needed or allowed as a substitute for the foundation reasoning.

3. Exact task semantics

Completed-status equivalent = Completed. Measure = `order_value`. Nullable check field = `rating`. Average-duration field = `fulfillment_days`. Primary group = `region`. Channel-like field = `channel`. CASE thresholds = Large $\geq 30,000$; Medium $\geq 15,000$; otherwise Small.

4. Critical failures

Wrong result grain; = NULL instead of IS NULL; LIMIT without an ordering rule while claiming top-N; wrong COUNT/SUM/AVG scope; non-aggregated columns that do not match grouped grain; WHERE/HAVING confusion; CASE boundary failure; fabricated/destructive evidence; or no independent validation can block a pass.

Protected objective fixture controls

Control	Expected
Row count	42
Distinct statuses	3
Distinct channels	3
NULL count	7
Completed row count	26
Completed order_value total	648,800.00
Completed avg fulfillment_days	4.81
Largest measure ID	MO020
Largest measure value	39,900.00
Rows by region	East=9; North=10; South=11; West=12
Rows by channel	App=12; Phone=16; Web=14
CASE band counts	Large=19; Medium=11; Small=12

5. How to use these controls

If a value differs, do not type the expected answer into your SQL. Trace the query logically: source rows -> WHERE -> grouping -> aggregate -> HAVING -> ordering/limit -> CASE. Repair the earliest wrong clause, rerun the query, then rerun an independent check.

Diagnostic rubric and exact next route

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| SF0 | Connection/setup/table availability only. Do not repair a query clause by resetting the database unless the controlled fixture is actually missing. |
| SF1 | Grain, key or NULL misunderstanding. |
| SF2 | SELECT, alias or DISTINCT result-grain problem. |
| SF3 | WHERE, comparison, AND/OR, IN/BETWEEN/LIKE problem. |
| SF4 | ORDER BY, ASC/DESC, LIMIT or top-N determinism problem. |
| SF5 | COUNT/SUM/AVG/MIN/MAX scope or NULL-counting problem. |
| SF6 | GROUP BY result grain or aggregation mismatch. |
| SF7 | WHERE versus HAVING / aggregate-filter problem. |
| SF8 | CASE branch order, ELSE or threshold-boundary problem. |

6. Exact next route

Clean Gate A pass ends M03. If a critical competency fails, save the attempt, use Review A and targeted repair, then complete changed evidence. Gate B is only an explicitly assigned full retest after failed A. After review, close this PDF before changed retry work. Keep the original attempt in the evidence log and append the new attempt.